

MUHAFIZ E WATTAN

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MUHAFIZ E WATTAN

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**A DISSERTATION SUBMITTED AS A PARTIAL FULFILLMENT OF THE REQUIREMENTS
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**DEPARTMENT OF COMPUTER SCIENCES
COMSATS UNIVERSITY ISLAMABAD,
WAH CAMPUS – PAKISTAN**

SESSION 2016-2020

FINAL APPROVAL

Certified that we have read this project report submitted by Muneer Hussain, Umar Shahzad and Irfan Ullah Khan, it is in my judgment, of sufficient standard to warrant its acceptance by COMSATS University Islamabad, Wah Campus, for the Bachelor degree in Computer Science.

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DEDICATION

We would like to dedicate this project to our parents and to our supervisor, as we are thankful for their guidance and help which was needed desperately.

Lastly, we would like to dedicate this project to the Pakistan Army Force and Pakistan police who have sacrificed their life there in the war on terror.

DECLARATION

We hereby declare that this document “MUHAFIZ E WATTAN” neither as a whole nor as a part has been copied out from any source. It is further declared that we have done this project with the accompanied report entirely on the basis of our personal efforts, under the proficient guidance of our teachers especially our supervisor Syed Faraz Ahmad.

ACKNOWLEDGEMENT

It is incumbent upon us to express our fervent gratitude to almighty ALLAH the most compassionate and merciful for giving us the courage to achieve this task in the field of software development study. We are indeed respectfully grateful to the Prophet HAZRAT MUHAMMAD (peace be upon him) who is forever a source of guidance and idea for all mankind.

We would like to express our immense gratitude to our instructor for his regular support and motivation that has encouraged us to come up with this project.

We are also thankful to our teachers and friends who have provide their whole hearted support at all times for the successful completion of our project Poultry Farming System. May GOD almighty Allah infuse us, with the energy to fulfill their gracious objectives and expectations and to further enlighten our competence!

PREFACE

In the preface of this final project report we would like to provide a brief introduction of MUHAFIZ E WATTAN Game final report. This final report of the project depicts all the phases & steps which we have taken to complete this project.

PROJECT BRIEF

PROJECT NAME: MUHAFIZ E WATTAN

PROJECT OBJECTIVE: The objective of this game is train counter terrorist forces with 3D virtual reality with a realistic 3D environment. It will save a lot of time, money and will also reduce accidents during training.

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COMPUTER USED: HP-LAPTOP

OPERATING SYSTEM: WINDOWS 10

TOOLS USED: UNITY 3D GAME ENGINE

ABSTRACT

This report explores the Research, Design and Implementation of 3D GAME features for making a 3D video game which based on the First-Person Shooting FPS, where the player takes role of the main character to kill enemies. This game is based on real world problem where a professional shooter fights with enemy in order to clearing the terrorist presence in the target area. It describes the process from storyboard to final production encompassing such areas as idea, design, modeling, texturing, rigging, mapping, sketching, rendering and Development. It will also cover the all aspects of tools which we used for creating the animated objects like Player, Enemy and Terrain. The game will deal with in artificial intelligence enemies, a good story board with some realistic characters and will solely be intended for single player and multi-player user.

TABLE OF CONTENTS

CHAPTER 1	1
INTRODUCTION	1
1.1 Introduction:	2
1.2 3D graphics	2
1.3 Impressive result	2
1.4 Graphical effects	2
1.5 Purpose of this Document	3
1.6 Intended Audience and Reading Suggestions	3
1.7 Scope of this Document	3
1.8 Overview	4
1.9 Story Abstract:	4
1.10 Level Design:	4
1.11 Intended Audience:	4
1.12 Main Focus of the Game:	4
CHAPTER 2	5
LITERATURE REVIEW	5
2.1 Overview	6
2.2 Business Context	6
2.3 Document Conventions	6
2.4 HISTORY	6
2.5 Design constraints	7
2.6 Standards Compliance	7
2.7 Hardware Limitations	7
CHAPTER 3	8
ANALYSIS	8
3.1 Introduction	9
3.2 General Descriptions	9
3.3 General Game Features	9
3.4 Similar System Information:	10
3.5 User Characteristics:	10
3.6 General Constraints:	10

3.7 Functional Requirements:	10
3.7.1 Player Models:	10
3.7.2 Object (character) Player spawn:	11
3.7.3 Menu:	11
3.7.4 Number of stages/levels	11
3.7.5 Camera Positions:	12
3.8 Non-Functional Requirements:	12
3.8.1 Playability:.....	12
3.8.2 Reliability:	13
3.8.3 Time-based animation.....	13
3.8.4 Security:	13
3.8.5 Reliability:	13
3.8.6 Preliminary Schedule	13
3.9 Hardware Requirements:.....	13
3.10 Developing Tools:.....	14
3.11 Scripting Language:	14
CHAPTER 4	15
DESIGN	15
4.1 Introduction:	16
4.2 Multi-platform	16
4.3 External Interface Requirements:	17
4.4 Layout of Application:	17
4.5 Level 1.....	19
4.6 Level 2.....	20
4.7 Level 3.....	21
4.8 Level 4.....	22
CHAPTER 5	23
UML DAIGRAMS	23
5.1 UML Diagram:	24
5.2 USE Case Diagram:	24
5.3 Sequence Diagram:	25
5.4 Class Diagram:	27
5.5 ACTIVITY Diagram:.....	30
CHAPTER 6	33

TESTING	33
6.1 Testing:.....	34
6.2 Aim of Testing:.....	34
6.3 Artifacts of Testing:	34
6.4 Test plan:	34
6.4.1 Traceability matrix:.....	34
6.4.2 Test case:	35
6.4.3 Test script:	35
6.4.4 Test suite:	35
6.4.5 Test data:	35
6.4.6 Test harness:	35
6.5 Unit Testing:	36
6.6 Limitations of Unit Testing:	36
6.7 Validation Testing:	36
6.8 Integration Testing:.....	36
6.9 Output Testing:	37
6.10 User acceptance testing	37
6.11 Black Box and White Box testing:	37
6.12 Usability feedback:	37
CHAPTER 7	39
SOFTWARE AND HARDWARE INTERFACE	39
7.1 Hardware:.....	40
7.2 Software:	40
7.3 Software and Hardware Interface:	41
CHAPTER 8	42
CONCLUSION	42
8.1 Conclusion:.....	43
8.2 Future works:.....	43
Appendix:	44
References:	44

CHAPTER 1

INTRODUCTION

1.1 Introduction:

“The MUHAFIZ e WTTTAN” is an action-based (VR) 3D first-person shooter game where the player takes role of the main character to destroy enemies. The gameplay is action-based with strategic elements. The only goal is to kill all enemies until no left before they kill our character.

The player plays from a first-person view in a (VR) 3D environment. The player experiences the game world through the eyes of a main character. This makes one feel that he/she is within the game. The player controls the main character. The abilities of our character as a player are walking and running in all four directions, crouching, jumping using-reloading-changing weapons.

The game has a design with both single-player and multiplayer mode. The player starts the game with three weapons and certain number of ammunition. During the game, the player has the ability to pick up more weapons and ammunition. By default, the player has one hundred health units. The attack of enemies reduces the health primarily. The effect of damage varies depending on the type of attack. The player has only one life. When his health reduces to zero or lower, the player will die. When our main character dies, the player loses the game. If the player kills all the enemies before the health reduce to zero or lower, the will wins the game/mission completed.

1.2 3D graphics

The game must contain 3D models, and render these in the game. 3D and VR environment is a requirement.

1.3 Impressive result

The game result must impress whoever plays the game. It should last long, and make the players return and play it over and over again.

1.4 Graphical effects

To achieve an impressive result, we would need to add modern graphical effects, such as real-time rendered soft shadows, motion blur, and ambient occlusion. The requirement for the game to contain 3D graphics introduced an interesting challenge for the project group, since all had none or little experience in 3D modeling. Spending time learning how to model proper 3D

models for the game it was therefore necessary. During the research we find out that what 3D modeling program to use, we found that we could use different studios to create models that we could later import to our game project. The game contains models made in Blender and used some unity assets. With these choices made, we created our environment sets to use unity 3D.

1.5 Purpose of this Document

The purpose of this document is to outline the functional and non-functional requirement of PC FPS multi-player game. This document also provides the requirements specification in detailed terms. This game (The Muhafiz e Wattan) covers all the necessary features which an FPS game should have such as AI opponent Enemy, different levels, interactions with the UI notes and above all damaging the player's health by attacking enemies.

1.6 Intended Audience and Reading Suggestions

We are going to make the SRS for our system. Particularly this SRS is intending for the developers, analyst, designer, coder, software engineer. So by the help of this SRS we can understand what we are going to do and what we are going to make. It shows the way to take the next step so we can say that it plays very important role in this software development. In this SRS first we are going to collect the requirements and then we do the planning that's how we are going to develop the game and after that we do analysis of this game and in the result of this analysis we get some specific requirements for the game so after that we will document those requirements.

1.7 Scope of this Document

This document describes all the requirement of the project. The purpose of this document is to provide virtual image of both structured and unstructured information. The Muhafiz e Wattan is an FPS multiplayer Game based on Virtual Reality (VR) for PC platform. This projects is for army personals which basic purpose will be training army personals through virtual reality. With the help of virtual reality soldier will be in a real world which will give them better understanding of the training and will know the training in advance which means what sort of activities will be taking place during counter terrorism operations. As now a days soldiers and law enforcement agencies personal are training and practicing operations technique through virtual reality because it save a lot of time and reduce the cost of training. This project will be

used by counter terrorism forces for training in 3D environment. The cost of this project is uncertain but can be estimated to \$700.

1.8 Overview

In this project we will take the requirements from army personals and according to those requirements we will implement it in Unity 3D gaming Engine and after playing this game army personals will have an idea how counter terrorism operation are covered and what kind of situations will occur and how to handle it .currently most of the western and advance countries are training their army personals through VR simulation/environment.

1.9 Story Abstract:

The MUHAFIZ E WATTAN is a virtual reality PC base First Person Shooter game. In this game a player will execute the AI opponent character (terrorist) with different Guns and in different places to stop him.

1.10 Level Design:

We design this episode of the game with different levels. In this way, one can find the details of the level design in the rest of the document.

1.11 Intended Audience:

The intended audiences for this gaming project are Army personals as well as teenagers, who would love to play such a games.

1.12 Main Focus of the Game:

- Each one of the different levels consists of goal that must be achieved.
- The player has to patrol around to kill the enemy and free the respectively area to complete the level.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview

This document provides an overview of the game functionality and interactions within the gaming system. Further this document also mentions the constraints and assumptions about the gaming system. The Muhafiz e Wattan is a PC base First Person Shooting game. In this game the player can switch to different levels in order to achieve the goals.

2.2 Business Context

Video games are more than just fun, they're big in business. The research firm Gartner predicts that video game sales top \$74 billion worldwide per year approximately and according to an estimate, it will reach \$112 billion by 2015 [6]. Video game development is a computer and mathematics-heavy field that spans the entirety of the many aspects involved in creating a video game. This includes everything from conception and storyline to graphic design and programming.

Our game is intended for PC based platform systems, which is abundantly used almost in all digital devices such as computers, laptops (for laptop and computer android emulator). This game targets army personals.

2.3 Document Conventions

This document is made according to the standards of IEEE. This is the low level requirement Document. The font style is Times new Roman, font size is 12, heading size is 14 and sub heading is 14.

2.4 HISTORY

Poor testing and bugs allocation are the ultimate mistake that leads to the failure of design. User sometime cannot be able to play games in a specific way.

For Example: The globally acclaimed Monster Hunter World was a complete failure. The world is delightful totally, yet the game is criticized for various nagging problems.

Scaling is the most common mistake that everyone does while designing. Forgetting to give each game a purpose or point to focus on. So, we are developing this game application, to make the information hiding simpler and user friendly. Our 3D game application is bit innovative than

others, in 3D game applications, user can easily understand and play any game in the initial stage within one or two attempts.

2.5 Design constraints

The system should be built by using a standard development tool like

- Visual Studio/ Mono developer (Program Developer)
- C-Sharp (Programming Language)
- Unity3d (Game Engine)
- Blander.
- Adobe Photoshop

2.6 Standards Compliance

Our system will follow the existing standards and regulations of fast-growing game industry.

2.7 Hardware Limitations

- 8 GB RAM
- Core i-5
- Windows 10.
- 3D graphics card.(GTX 1050Ti)

CHAPTER 3

ANALYSIS

3.1 Introduction

“The MUHAFIZ E WATTAN” is an action-based 3D (VR) first-person shooter game [23]. Where the player takes role of the main character to destroy enemies. The gameplay is action-based with strategic elements. The only goal is to kill all enemies until even one is not left before they kill our character.

The player plays from a first-person view in a 3D VR environment. The abilities of our character as a player are walking and running in all four directions, crouching, and jumping, reloading-changing weapons. The game has a design with both Single-player and Multi-player mode. The player starts the game with weapon and certain number of ammunition. During the game, one has the ability to pick up more weapons and ammunition. By default, the player has one hundred health units. The attack of enemies reduces the health primarily. The effect of damage varies depending on the type of attack. However, the player has only one life. When his/her health reduces to zero or lower, one will die. When our main character dies, the player loses the game. If the player kills all the enemies before the health reduce to zero or lower, one wins the game.

3.2 General Descriptions

Product Functions

- The FPS game the user will be able to start and accomplish their mission by locating and eliminating the enemy.
- The player will kill the enemy and pick the certain object in order to complete the level.
- This feature is also can accomplished in multiplayer and VR (virtual reality) mode.

3.3 General Game Features

This FPS game, named “MUHAFIZ E WATTAN”, involves:

- FPS player.
- Three different levels.
- Enemy (terrorist) as an AI.
- VR

- Multiplayer gaming mode.

3.4 Similar System Information:

"MUHAFIZ e WATTAN" is a first-person (VR) game, providing dynamic, clearing the area from enemy to complete levels. In each level we are going to reach to the required goal (clearing area) which will lead it towards the completion of that level and it will then trigger it to the next level.

3.5 User Characteristics:

There is only one type of user that is the player which interacts with the system. The user will be able to start level. User will be able to pause, resume and quit the level during a play mode. User can also be directed back to main menu from pause menu screen.

3.6 General Constraints:

- Standalone not online in case of local network.
- Initially work on 3 levels.

3.7 Functional Requirements:

1. Game should have a player character (FPS) [26]
2. Player will be respawned after getting zero health.
3. Game have a User Interface menu.
4. Game have a certain number of stages/levels.
5. One Camera Position in the game.

3.7.1 Player Models:

3.7.1.1 Description:

Game have a one playable character for the user who is being provided with the rifle in single player mode and player can be multiplayers in multiplayer mode.

3.7.1.2 Criticality

The main critical object in the game is FPS (First Person Shooter) and Multiplayer Networking Which handle multiple player at the same time and AI enemy character as well.

3.7.1.3 Cost and schedule

Adding more enemies and building level costs more time in adjusting it and make it work properly.

3.7.2 Object (character) Player spawn:

3.7.2.1 Description

Object player will be spawn at the spot defined in the script, after getting crash by enemy AI opponent terrorist.

3.7.2.2 Criticality

It is a main character of the game, all the game depends on this character and full game was resolve around this character which was controlled by the player to accomplish the mission.

3.7.3 Menu:

3.7.3.1 Description

Game have the menu list in the start of the game and another menu list is also shown during game play when player press the “ESCAPE” key by which the player was able to resume, direct to main menu or exit game.

3.7.3.2 Criticality

It is one of the most critical requirements because with a menu, player can interact with game easily rather than a command line console.

3.7.3.3 Technical Issue

Input device and user inputs.

3.7.4 Number of stages/levels

3.7.4.1 Description

The player is going to reach specified score in order to switch to different level, or selecting level from start .The number of levels is 4 including multiplayer level.

3.7.4.2 Criticality

A good First-Person Shooting game needs attractive levels to increase playing interest.

3.7.4.3 Dependencies with Other Requirements

Menu

Storyboard

3.7.5 Camera Positions:

3.7.5.1 Description

Player will have two types of camera positions, main screen camera and navigation camera.

3.7.5.2 Criticality

it is also a critical requirement because with it user can control player waking and running.

3.7.5.3 Technical issue

Input device malfunctioning

3.8 Non-Functional Requirements:

Following are some of the major non-functional requirements essential for our gaming project:

3.8.1 Playability:

In the game reviews and evaluations, the game is assessed generally in terms of four categories: game play, atmosphere, video and sound, while game play is considered as the most important feature of a game. According to this fact, the playability of the game is a major concern and requirement. The other features of the game are to be designed accordingly, with the ultimate aim to increase playability, thus player satisfaction.

3.8.2 Reliability:

The game has highly distributed data operations, and for that reason a strong control of consistency and synchronization is to be achieved. This stands to be a very important topic in system robustness and thus user satisfaction.

3.8.3 Time-based animation

The game algorithms would be infallible regardless of a faster-than-expected or slower-than-expected frame-rate.

3.8.4 Security:

Game will be secured on the basis of its scripts and will have restriction for entering into the basic scripts of the game.

3.8.5 Reliability:

The system would be reliable and have no problem with respect to functionality.

3.8.6 Preliminary Schedule

The system is divided into six parts that are as under Main menu

- Terrain 3D
- Coding of the system
- Player object and Enemy AI Opponent
- Animations of Enemy AI.
- Virtual Reality (VR).
- Documentation of the system.

3.9 Hardware Requirements:

The Muhafiz e Wattan is a PC based game which will be applicable soon on PC. The expected size of the game will be less than 6GB and dual core processor PC to run the application smoothly. There should be at least 500 MB free space on hardware, due to future updates and it will need GTX 1050Ti 3d graphics cards.

3.10 Developing Tools:

The Working tools and software are:

- Unity 3D 2017 and above
- Blender[6]
- Star UML (for sequence and USE CASE diagrams)
- Adobe PHOTOSHOP.
- Adobe ILLUSTRATOR.

3.11 Scripting Language:

C# [4] (pronounced "C-sharp") is an object-oriented programming **language** from Microsoft that aims to combine the computing power of C++ with the programming ease of Visual Basic. **C#** is based on C++ and contains features similar to those of Java.

CHAPTER 4

DESIGN

4.1 Introduction:

There are a number of issues you must bear in mind before embarking down the path of PC application design. This section describes how the final design came around. The designs include reason and thought behind the user interface, general architecture, choice of tools, and platform choice. Design planning is important to ensure that incorrect and unnecessary features are not implemented. Doing so is time consuming and thus is why this stage is important in the project. This stage ensures that ideas for the project gathered from background research and problem analysis are feasible in 20 weeks. This will help direct the project to provide potential ideas for the problems listed earlier. The following sub chapters listed here represent a single feature or idea and are an aim to solve part of the problem.

3D Viewer:

The Scene View has a set of navigation controls to help you move around quickly and efficiently. The Scene Gizmo is in the upper-right corner of the Scene View. This displays the Scene View Camera's current orientation, and allows you to quickly modify the viewing angle and projection mode.

The Scene Gizmo has a conical arm on each side of the cube. The arms at the forefront are labelled X, Y and Z. Click on any of the conical axis arms to snap the Scene View Camera to the axis it represents (for example: top view, left view and front view). You can also right-click the cube to bring up a menu with a list of viewing angles. To return to the default viewing angle, right-click the Scene Gizmo and click Free.

4.2 Multi-platform:

Create the program on more than one platform to expand the market audience for the application. Being platform exclusive has the advantage that you only need to develop one application which makes maintaining and support easier as well as the initial development cost and time. This could be difficult to implement. The options are to either use a framework which won't be able to take advantage of the platform's user interface guidelines and adaptors or native development for each application will be required which would mean interface code would have to be recreated for each

4.3 External Interface Requirements:

The game display main menu with one start button, setting button and quit. User will start game with level 1 then after its completion user will be able to move to the next level.

4.4 Layout of Application:

The part of graphic design that deals in the arrangement of visual elements on a page. It generally involves organizational principles of composition to achieve specific communication objectives. The high-level page layout involves deciding on the overall arrangement of text and images, and possibly on the size or shape of the medium. In fig 4.1, Selection Menu panel contain a Play, Setting, Help and Quit buttons. Play button will direct you to selection Mode Menu as shown in fig 4.2.

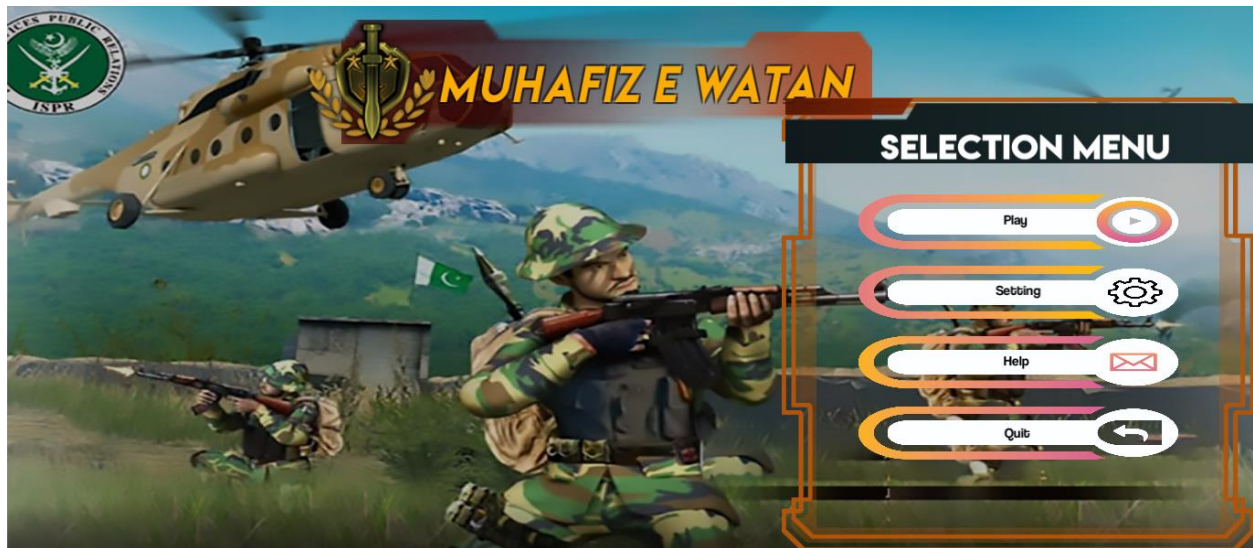


Figure 4.1(Main interface)

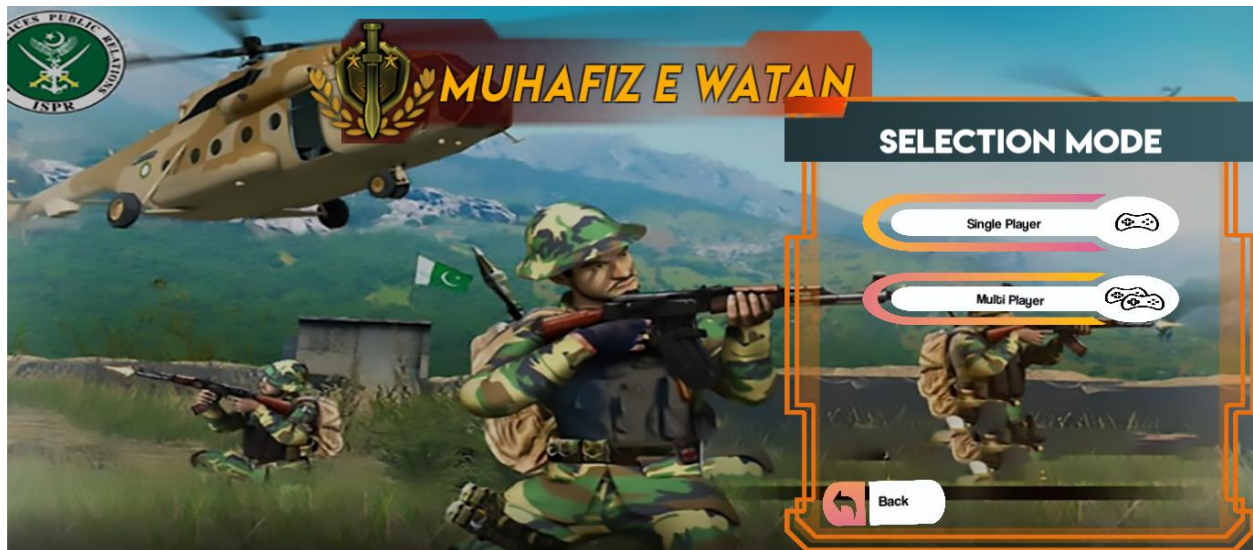


Figure 4.2(Selection Mode)

In the Selection mode (fig 4.2) shows three buttons one is Single Player button and second is Multi player button and the third one is back button. If Player will Select Single player button they will direct to Level selection Mode Menu as shown in fig 4.3.

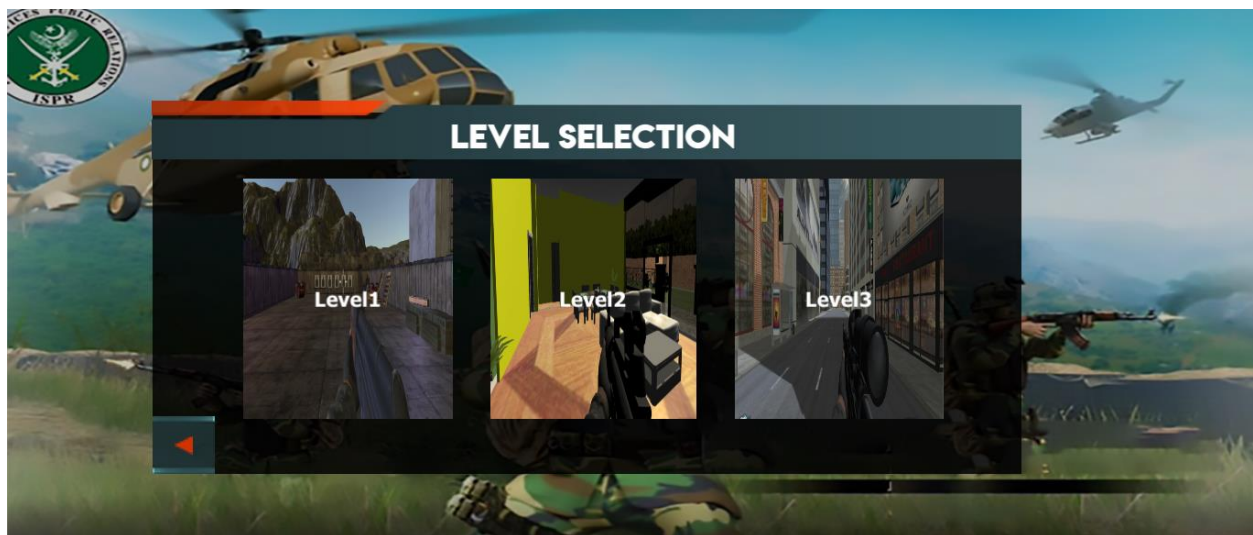


Figure 4.3(Level selection)

.In the level selection (fig 4.3) you have three option level1, level2 and level3.If you have select level1 then you will be directed to level1.

4.5 Level 1.

In fig 4.4, the whole structure of scene is being displayed (level 1). In fig 4.4, the 2nd display view of level 1. In fig 4.5, safe place is shown from where actually the game starts. Level 1 is a Demo level .The main objective of level 1 is to train the army personals that how to target an enemy by using target object.



Figure 4.4(Single Player)

The main objective of level 1 is that the Player will able to target all the targeted objects which are place in the scene. Level 1 is basically a training demo scene which is used to train the players. There are 12 targeted objects in the scene. Player will target an object one by one.

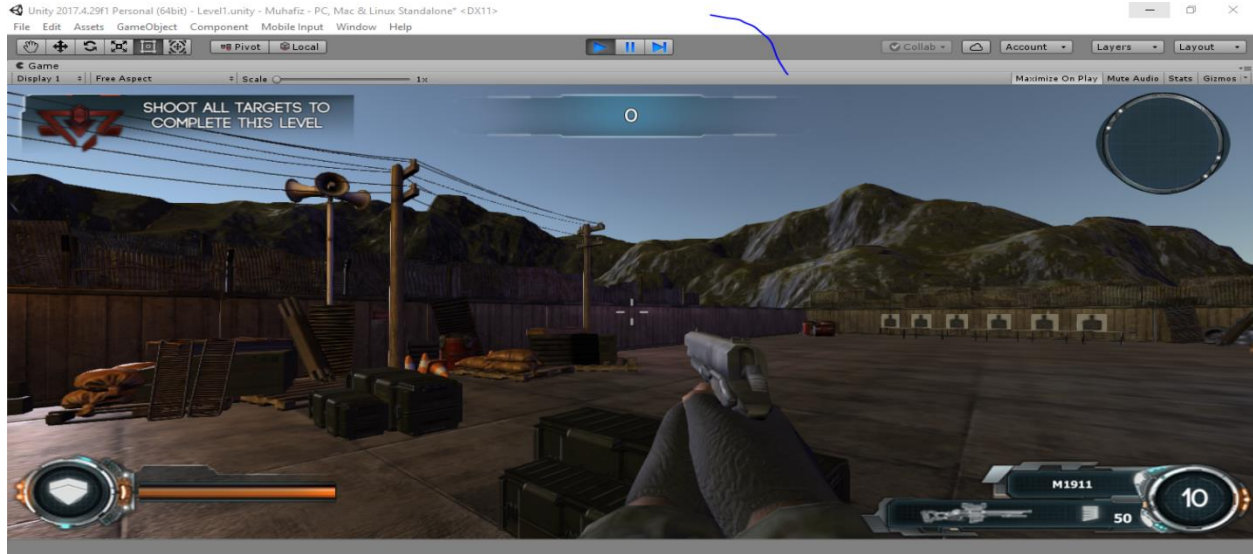


Figure 4.4(Single Player)



Figure 4.5(Single Player)

.In the level selection (fig 4.3) you have three option level1, level2 and level3.If you have select level 2 then you will be directed to level 2.

4.6 Level 2.

In fig 4.6, the whole structure of scene (level 2). In this scene the player takes role of the main character to destroy enemies. The gameplay is action-based with strategic elements. The only goal is to kill all enemies until no left before they kill our character. The main interface of scene is given below fig 4.6.



Figure 4.6(Single Player)

Pause Menu panel contain a Restart, Resume, and Main Menu buttons. Resume button will direct you to selected level, Restart button will play game again and Main Menu button will direct to main user interface.

The pause menu during the gameplay in fig 4.7.

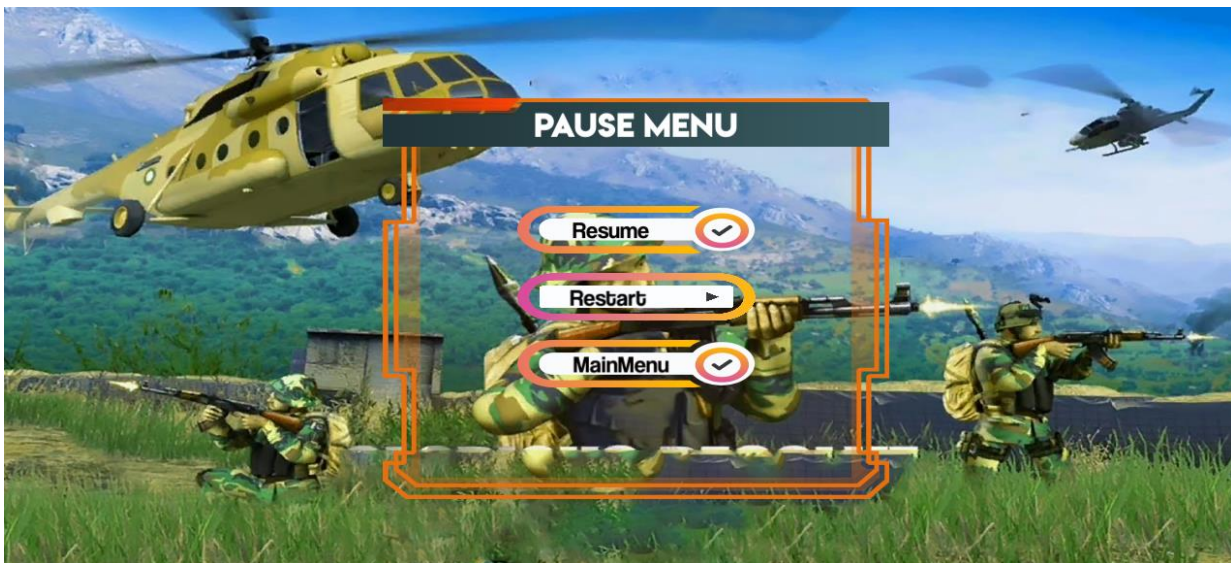


Figure 4.7(User Interface)

In the level selection Mode (fig 4.3) you have three option level1, level2 and level3.If you have select level 3 then you will be directed to level 3.

4.7 Level 3.

In fig 4.8, the whole structure of scene (level 3). In this scene the player takes role of the main character to destroy enemies. The gameplay is action-based with strategic elements. The only

goal is to kill all enemies until no left before they kill our character. The main interface of scene is given below fig 4.8.



Figure 4.8(Single Player)

In the Selection mode (fig 4.2) shows three buttons one is Single Player button and second is Multi player button and the third one is back button. If Player will Select Multi player button they will direct to level 4.

4.8 Level 4.

In fig 4.9, the whole structure of scene (level4).Multiplayer.



Figure 4.9(Multiplayer)

In this scene the player takes role of the main character to destroy other character. The gameplay is action-based with strategic elements. The only goal is to kill all others players until no left. The main interface of scene is given below fig 4.9

CHAPTER 5

UML DAIGRAMS

5.1 UML Diagram:

The Unified Modeling Language (UML) is a general-purpose, developmental, modeling language in the field of software engineering that is intended to provide a standard way to visualize the design of a system.

5.2 Class Diagram:

A class diagram in the Unified Modeling Language is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations, and the relationships among objects fig. 5.2.

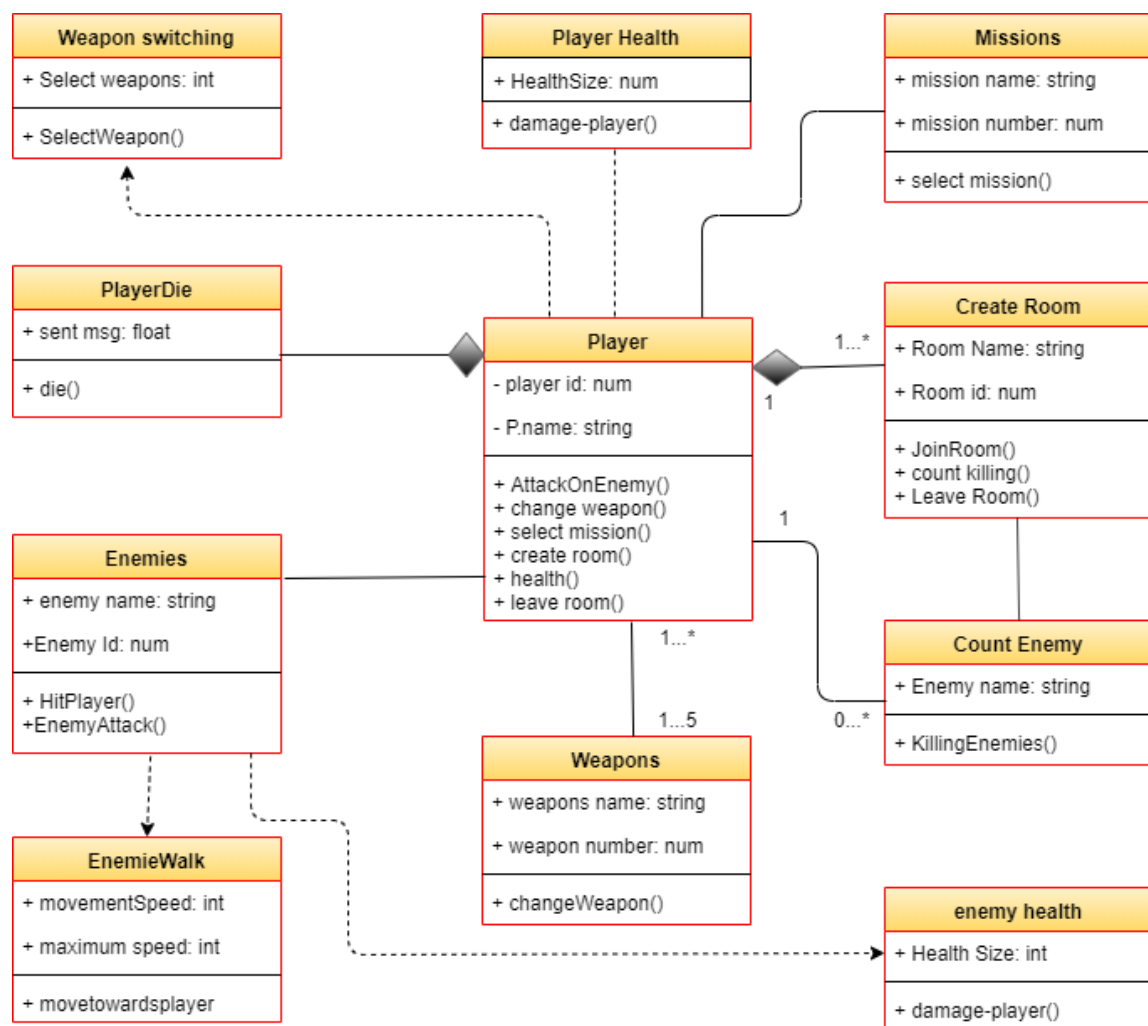


Figure 5.2 Class diagram

There are 11 classes. Which are listed below.

1. Player:

Player class have two attributes (Int : Player-Id, string Player Name) and have six operations(Attack On Enemy, Change Weapons, Walking , Jumping, Selection Missions, Health).In this class diagram player class show the relationship between the other classes like (player die ,Weapons class ,etc.).

2. Enemy:

Enemy class have two attributes (Int: Enemy-Id, string Enemy Name) and have three operations (Hit Player, Enemy Attack, walking).In this class diagram Enemy class show the relationship between the other classes like (Enemy walk, and player class).

3. Enemy Walk:

Enemy Walk class have two attributes (Int : movement speed, maximum speed) and have one operations(move toward player).In this class diagram Enemy Walk class show the relationship between the other classes like (Enemy) enemy walk is inherit from enemy class.

4. Player Health:

Player class have two attributes (Int : player-Id, string Player Name) and have six operations(Attack On Enemy, Change Weapons, Walking , Jumping, Selection Missions, Health).In this class diagram player class shows the relationship between the other classes like (player die ,Weapons class ,etc.).

5. Weapons:

Weapons class have two attributes (Int: weapons number, string weapons Name) and have one operations (Change Weapons).In this class diagram Weapons class shows the relationship between the other classes like (player class).

6. Missions:

Missions class have two attributes (Int: mission-Id, string mission Name) and have one operations (Select Mission).In this class diagram player class shows the relationship between the other classes like (player).

7. Create Room:

Create Room class have two attributes (Int : Room-Id, string Room Name) and have three operations(Join Room , Count Killing, Leave Room).In this class diagram create room class shows the relationship between the other classes like (Count Enemy class).

8. Enemy Health:

Enemy Health class have one attributes (Int: health size) and have one operations (damage player).In this class diagram player class shows the relationship between the other classes like (Enemy class).

9. Player Die:

Player Die class have two attributes (Int: sent message, string Player Name) and have one operations (Die).In this class diagram player die class shows the relationship between the other classes like (player).

10. Weapons Switching:

Weapons Switching class have one attributes (Int: selection weapons) and have one operations (selection weapons).In this class diagram weapons switching class shows the relationship between the other classes like (player, Weapons class).

11. Count Enemy:

Count Enemy class have one attributes (string: Enemy Name) and have one operations (Killing Enemies).In this class diagram player class shows the relationship between the other classes like (create room).

5.3 USE Case Diagram:

In fig 5.3 the use case diagram represents the overall scenario of our system. A scenario is a sequence of steps describing an interaction between a User and system. Represents that user gives sample of The Muhafiz e Wattan Game. Which tell us how the user interacts with their system.

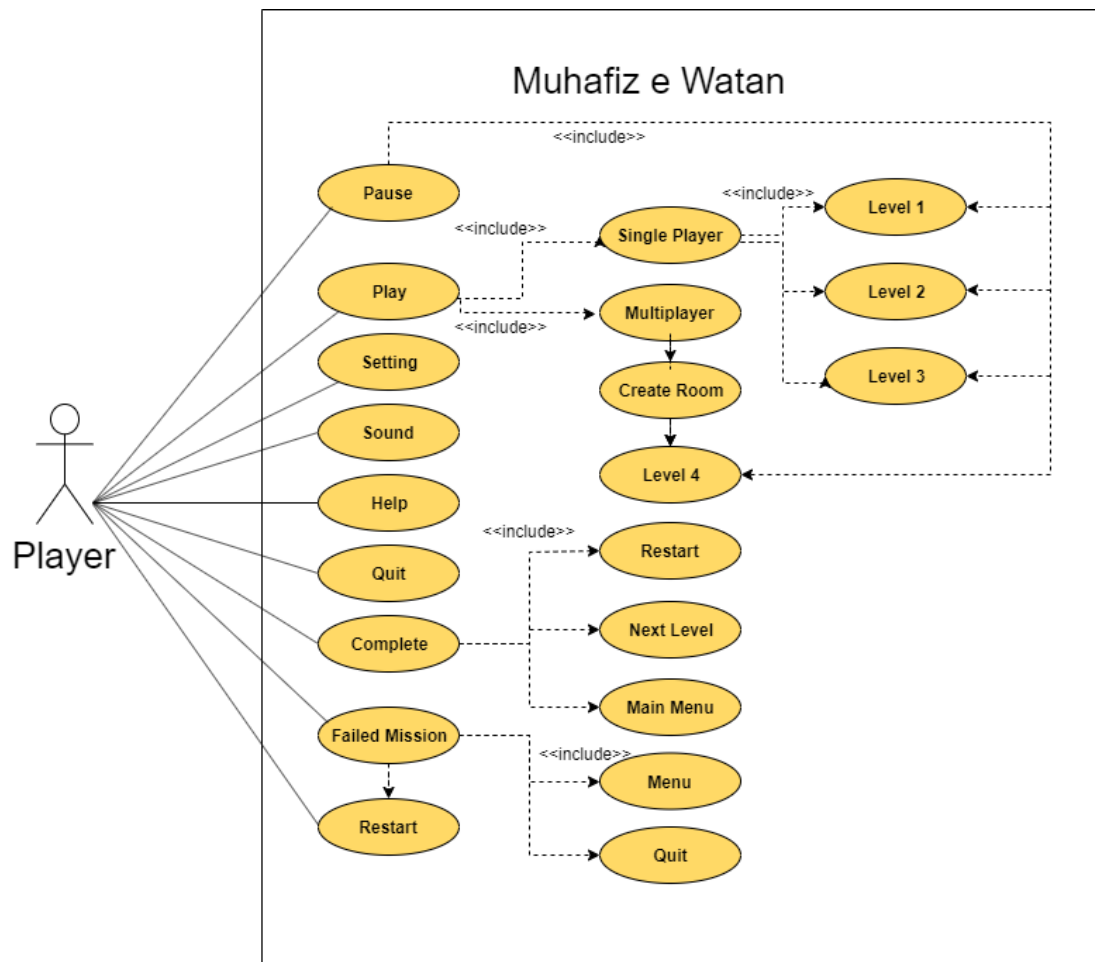


Figure 5.3 Sequence diagram

There are 21 Use Cases in our project and Actor is our project. Which are listed below.

1. Player: Player is an actor who interact with use cases.
2. Play: It is a use case and it works to play game (Multi Players and First-person)
3. Setting: It controls the overall game like the sound system and game levels difficult, medium easy.

4. Sound: It is a use case to control the game volume.
5. Help: this use case used for help when we don't know about the game (How to playing, level setting, and sound setting)
6. Quit: This use case used for when we close our game or any mission.
7. Complete: When our mission completes then this use case appears on the screen
6. Failed Mission: When our mission failed then we used this use case.
8. Restart: When we start our mission from the start then we used this use case.
9. Create Room: It is used for multi players when we create rooms for different players.

5.4 Sequence Diagram:

A sequence diagram shows object interactions arranged in time sequence. It depicts the objects and classes involved in the scenario and the sequence of messages exchanged between the objects needed to carry out the functionality of the scenario.

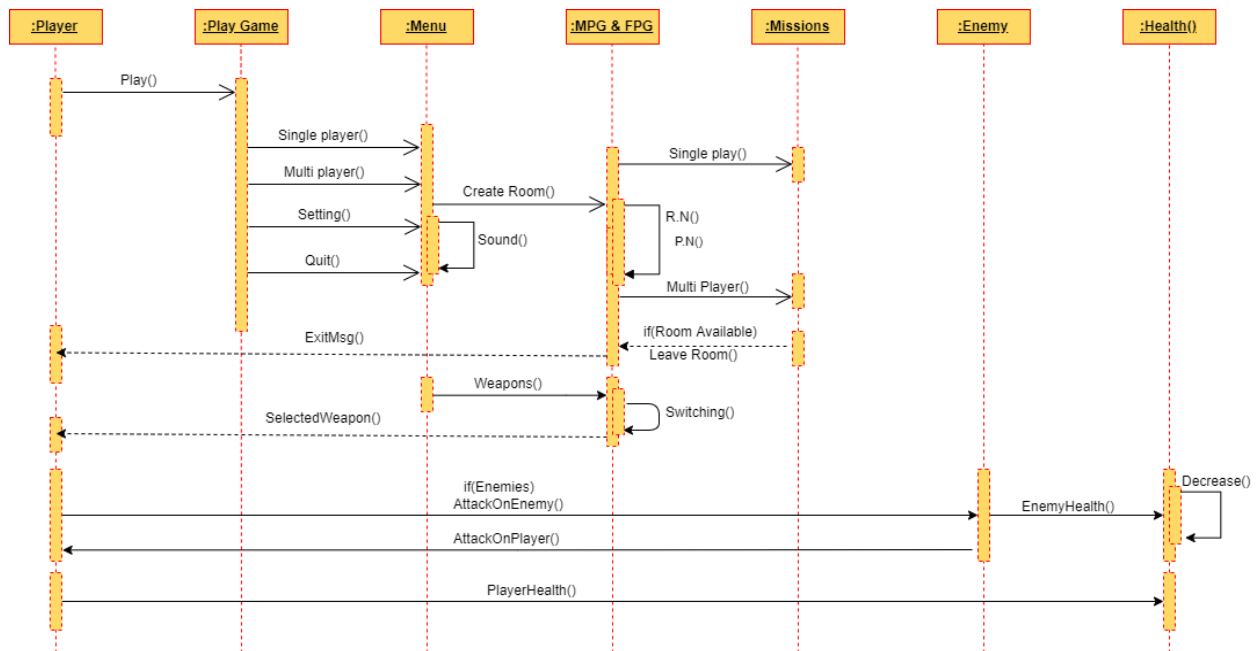


Figure 5.4 Sequence diagram

1. Player: It is an object of the player class and the basic function of this object is to control all over the game.
2. Play Game: It is an object and used for multiple purposes like single-player, multi-player and game setting, etc.
3. Menu: Through this object, we create rooms for multi-players to control sounds and change weapons.
4. MPG & FPG: It is used for when we play a game then first we choose which option that we played single-player or multi-player.
5. Missions: Basically it is an object of the level class that we select in which level we want to play the game.
6. Enemy: It is the object of an enemy class and we handle enemies and their health in this object.

7. Health: We handle the enemy and player health in this object.

5.5 ACTIVITY Diagram:

Fig 5.5 is Activity diagram, used to graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency of our system.

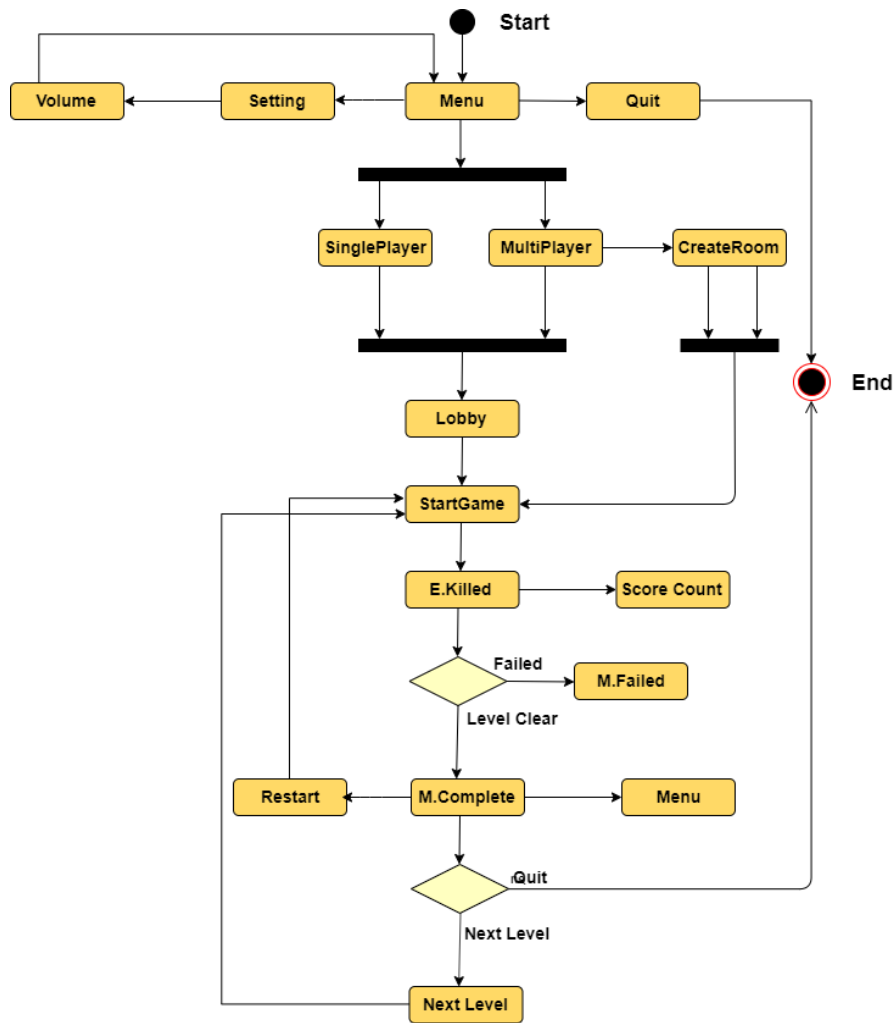


Figure 5.5 Activity diagram

1.1. Setting: After the Main menu, our next activity is “Setting” and in this activity, we handle the sound of the game.

1.1.1. Volume: In this activity, we handle the volume.

1.2. Quit: In Main Menu the last activity is “Quit” and we close our game in this activity.

2. Single Player: This activity is used for the first-person player.

3. Multi-Player: This activity we used when played more than one person.

3.1. Create Room: When we playing game more than one person (Multi-Players) than we need first to create room (Player ID and Room name).

4. Lobby:

5. Start Game: This activity is used when our game is in play mode for first-person and also used for multi-players.

6. E.Killed: It is used for enemy killing during playing.

6.1. Score Count: When player fire on enemy and enemy kills then this activity is used.

7. Mission Failed: If a player dies than using this activity.

8. M.Complete: When we complete our target in any mission then we used this activity to go for the next level, restart mission or close our game.

8.1. Restart: During playing game when we click on this activity then our mission is displayed from start.

9. Next Level: When one mission is complete and start the next level than we used this activity.

10. End: When we close our game then using this activity.

CHAPTER 6

TESTING

6.1 Testing:

Testing defines the status of the working functionalities of any particular system. Through testing particular software one can't identify the defects in it but can analyze the performance of software and its working behavior. By testing the software we can find the limitations that become the conditions on which the performance is measured on that particular level. In order to start the testing process the primary thing is requirements of software development cycle. Using this phase the testing phase will be easier for testers.

The functional testing, which defines the specified function of a particular code in the program. This type of testing gives us a brief description about the programs performance and security in the various functional areas.

6.2 Aim of Testing:

The main aim of testing is to analyze the performance and to evaluate the errors that occur when the program is executed with different input sources and running indifferent operating environments. In this project, I developed a 3D games application based on Unity which focuses on FPS and Multiplayer mode. The main aim of testing in this project is to find the compatibility issues as well as the working performance when different sources are given as the inputs.

6.3 Artifacts of Testing:

For testing the application the testing process produces several art facts. The different art facts are:

6.4 Test plan:

Test plan of this project was that it should be tested form the beginning to the end every part/task that had been completed after completion that component/function is tested properly after testing then the next component/function of the product development had commenced .The testing plan was that it should be tested form building to the very end of the project.

6.4.1 Traceability matrix:

A traceability matrix is a table that link the design documents to the text documents. This is also changes the test processes when the source documents are changed .The files of the project is

arranged in proper order so that the tester and developer have not difficulty to trace any change and upgrading/updating the project.

6.4.2 Test case:

Test case for all valid inputs.

Purposes	To check inputs of the game.
Screen to tested	Gameplay and GUI
Input	Invalid input
Output	No action preformed.
Expected Result	Game is perfectly in working condition.
Actual result	Same is as expected result

6.4.3 Test script:

A test script is a product of work generated by automated regression tools. It is a combination of test case, test procedure and test data.

6.4.4 Test Suite:

A test suite is a collection of test cases. The test suite consists of the detailed instructions and the role of each collection of test cases.

6.4.5 Test data:

The place where the test values and components can be modified is known as Test data.

6.4.6 Test harness:

A test harness is a collection of software, inputs, outputs and configurations used for the application. There are different types of approaches for testing an application. The types of testing are

- Unit testing
- Validation testing

- Integration testing
- User acceptance testing
- Output testing
- Black box and white box testing

6.5 Unit Testing:

We have test every single component of this project for example GUI (graphical user interface) button and menu and every single functions like shooting, walking and gameplay sounds.

6.6 Limitations of Unit Testing:

Unit testing is limited to test only the functionality of the units. It can't identify integration errors, performance problems and system problems. Unit testing can show the errors which occur in the units when the testing runs. It may not display the errors that currently are absent.

6.7 Validation Testing:

We have validated the final product of this project everything work perfectly and meet the needs and requires which were written in the SRS.

The primary objectives of testing during the requirements phase are to:

- Determine that the requirements fairly represent what the is needed
- Determine that the needs have been defined and documented
- Verify that a reasonable and valid cost/benefit study has been performed
- Determine that the business problem has been solved
- Verify that the control requirements have been specified
- Verify that a reasonable process was followed in developing the business solution.

6.8 Integration Testing:

There are basically main three features of Muhfiz e Wattan .The first one is single player 2nd one is multiplayer and the 3rd one is VR (virtual reality) mode .so we have test every single feature of this project and after testing we have integrated all the features and then again we have

performed integration testing by combining all these features and tested the project as a whole single product.

- What will be tested for example software features.
- What is the time scale and time management?
- Responsibilities, e.g. personnel.
- Testing pass and fail condition.
- Risk involved.
- Approval from all important people involved.

6.9 Output Testing:

The output of muhfiz e wattan is according to the expectations of user. Every function and action of the project had an expected output which the users have desired for example during gameplay (when playing in game mode) or off gameplay (Menu, sounds and button) everything has a desired output as the user desired.

6.10 User acceptance testing:

User acceptance test had been performed by user/supervisor and they have agreed that the product is according to the user need and requirements. The user has tested all the three features of the game and is functional working properly.

6.11 Black Box and White Box testing:

Black box testing is performed by the user by playing the muhfiz e wattan game and all the three mode (single player, multiplayer and VR mode) which work completely fine and the White box is carried out by the developer everything is perfect from design to coding and implementations there is no bugs or errors in the product.

6.12 Usability feedback:

A user study was taken to gather user opinion about the ease of use of the application, the study's goal is to show how well this application has met requirements. It is important to get a view of the application other than the developers, users may pick up on usability issues that a developer wouldn't notice. This feedback was done through participants testing the

application and answering a simple questionnaire about the usability of the system. This feedback is different to the observation done in design as that was aim to improve the system whereas this is evaluating it. The tests consist of the participant creating a room, publishing it, and finding their room.

CHAPTER 7

SOFTWARE AND HARDWARE INTERFACE

7.1 Hardware:

The overall design of our Virtual Reality (VR) FPS game in terms of Modern computer games place great demand on the computer's hardware, often requiring a fast-central processing unit (CPU) to function properly. CPU manufacturers historically relied mainly on increasing clock rates to improve the performance of their processors, but had begun to move steadily towards multi-core CPUs by 2011. These processors allow the computer to simultaneously process multiple tasks, called threads, allowing the use of more complex graphics, artificial intelligence and in-game physics.

Similarly, 3D games often rely on a powerful graphics processing unit (GPU), which accelerates the process of drawing complex scenes in Real-time.

7.2 Software:

Fig 7.2 shows that in term of software, the player can be play the game while using oculus rift s. Diagram show the interaction of player and system.



Figure 7.2

Fig 7.2 show the interaction between the player and the computer. The human body shows how to used oculus rift s. Also the connective between desktop and CPU.

7.3 Software and Hardware Interface:

The basic hardware modules are oculus rift s, mouse, keyboard and VGA screen, and therefore the interface is to connect this four modules with software. To achieve this, we should implement this on software and hardware drivers. Some main hardware are shows below.



OCULUS RIFT S: is a virtual reality headset device. Which is use to view 3D environment in virtual reality.

KEY BOARD: is a input device .which is used to move player in all four directions (forward, backward, left and right) and also used to change the weapons in game mode.

MOUSE: is a also input device. Which is used to firing in game mode and also used to change Weapons.

VGA SCREEN: is an output device. Which is used to display game on screen.

JOYSTICK: is a input device .which is used to move player in all four directions (forward, backward, left and right) and also used to change the weapons in game mode.

CPU: Central processing Unit.

CHAPTER 8

CONCULUSION

8.1 Conclusion:

The main purpose of this game is to “Train Anti-Terror Personals ” and the objective of the game is to practice ,military tactics, techniques and strategy of military, which law enforcement agencies will to combat are prevent terrorism. Terrorism is a part of a broader insurgencies which our country is currently facing. It will help anti -terror forces to employ counter insurgency measure.

The process we used is well fitted to projects that need fast specified results. We believe that our working method enables us to achieve more features in our game than if we used a more pre-planned method. With our method, developers can get started directly with game specific features, and do not need to think about structure and writing general code. For larger projects, we think that planning and having a good structure, becomes even more important.

8.2 Future works:

When implement all those basic functions stated above, we can focus on following.

- Improving the other aspects of this shooting Game.
- We would like to add music, video and some audio effect and more type of enemy to make this game feel more realistic.
- We can improve the play-ability by implementing some extra characters.

Appendix

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